

Air Quality and Attributable Health Burden Evaluations Through Long-Term AQI Analysis (1980–2025)

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Abstract

This study evaluates long-term air quality trends from 1980 to 2025 using annual Air Quality Index (AQI) data. Exploratory data analysis, AQI category distributions, median AQI trajectories, and non-parametric trend estimation were applied to distinguish episodic pollution events from chronic air quality degradation. Results demonstrate a statistically significant upward trend in median AQI, with Sen’s slope estimating an increase of approximately 0.30 AQI units per year (95% CI: 0.21–0.39). Distributional analysis reveals a sustained decline in “Good” AQI days and a corresponding increase in “Moderate” AQI days, while Unhealthy and more severe categories remain rare. Evidence from AQI heatmaps, median trends, and stacked distributions consistently indicates a gradual shift toward chronic moderate pollution exposure rather than increased frequency of extreme events. Population attributable risk analysis suggests that this shift is associated with a growing long-term public health burden, particularly for sensitive populations. These findings highlight the need for air quality management strategies that target baseline pollution reduction rather than focusing solely on extreme events.

Keywords: Air Quality Index, AQI trends, Sen’s slope, Mann–Kendall test, population attributable risk, PM_{2.5}, Ozone, Environmental health

1 Introduction

Air quality is a critical determinant of population health. While regulatory interventions have reduced the frequency of extreme pollution episodes, long-term exposure to moderate air pollution may produce substantial cumulative health effects. This study examines long-term AQI behavior to determine whether observed air quality changes reflect episodic

pollution events or a structural transition toward chronic exposure. By combining exploratory data analysis with robust non-parametric trend estimation and health burden attribution, this analysis provides an integrated assessment of both environmental and public health risk.

2 Data

This study uses annual air quality and asthma morbidity data for Monroe County, New York. Air pollution measures, including AQI statistics and pollutant indicators (PM_{2.5}, ozone, CO, NO₂, PM₁₀), were obtained from the Centers for Disease Control and Prevention (CDC) National Environmental Public Health Tracking Network. Asthma hospitalization and emergency department visit rates (2000–2023) were sourced from the New York State Department of Health (NYSDOH) asthma surveillance reports. Additional asthma prevalence context was referenced from the CDC Behavioral Risk Factor Surveillance System (BRFSS) and CDC Vital Signs reports. These datasets were integrated at the annual level to support analysis of exposure-response relationships.

3 Methods

3.1 Exploratory Data Analysis

Exploratory data analysis was conducted to characterize both intensity and frequency changes in air quality over time. Methods included annual AQI category heatmaps, time series analysis of median AQI, and stacked bar charts of AQI category proportions.

3.2 Trend Analysis

Long-term trend magnitude was estimated using Sen’s slope:

$$Q = \text{median} \left(\frac{x_j - x_i}{j - i} \right), \quad j > i \quad (1)$$

Trend significance was evaluated using the Mann–Kendall non-parametric test, which is robust to non-normality, missing values, and outliers.

3.3 Confidence Intervals

Confidence intervals for Sen’s slope were computed using Kendall score variance and normal approximation.

4 Health Burden Attribution

To quantify public health implications, Moderate AQI exposure was defined as the annual proportion of days classified as “Moderate”. Population attributable fraction (PAF) was calculated as:

$$\text{PAF} = \frac{P_e(RR - 1)}{P_e(RR - 1) + 1} \quad (2)$$

where P_e represents exposure prevalence and RR denotes relative risk associated with chronic moderate air pollution exposure. Conservative literature based risk estimates were used to avoid overstatement.

5 Results

5.1 AQI Category Distribution Shifts

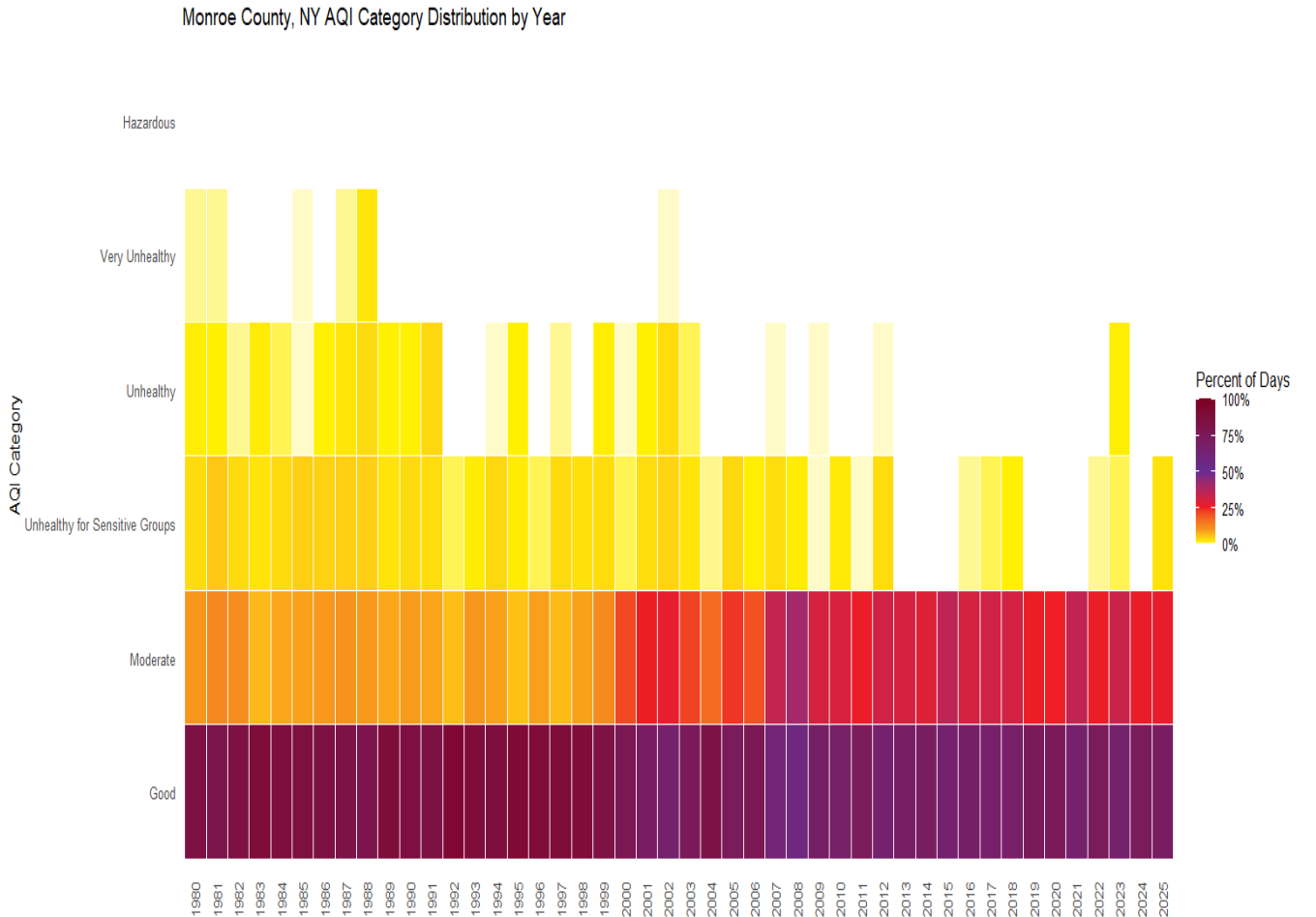


Figure 1: Annual AQI category distribution heatmap for Monroe County (1980–2025).

The heatmap reveals that “Good” AQI days dominate throughout the study period, particularly prior to approximately 2000. After 2000, “Moderate” AQI days increase

steadily, becoming a substantially larger fraction of annual exposure. “Unhealthy for Sensitive Groups” days appear intermittently after 2000 and increase modestly post-2005. Unhealthy and Very Unhealthy days remain rare, and Hazardous days are virtually absent. This pattern indicates gradual air quality degradation without persistent extreme pollution events.

5.2 Median AQI Trend

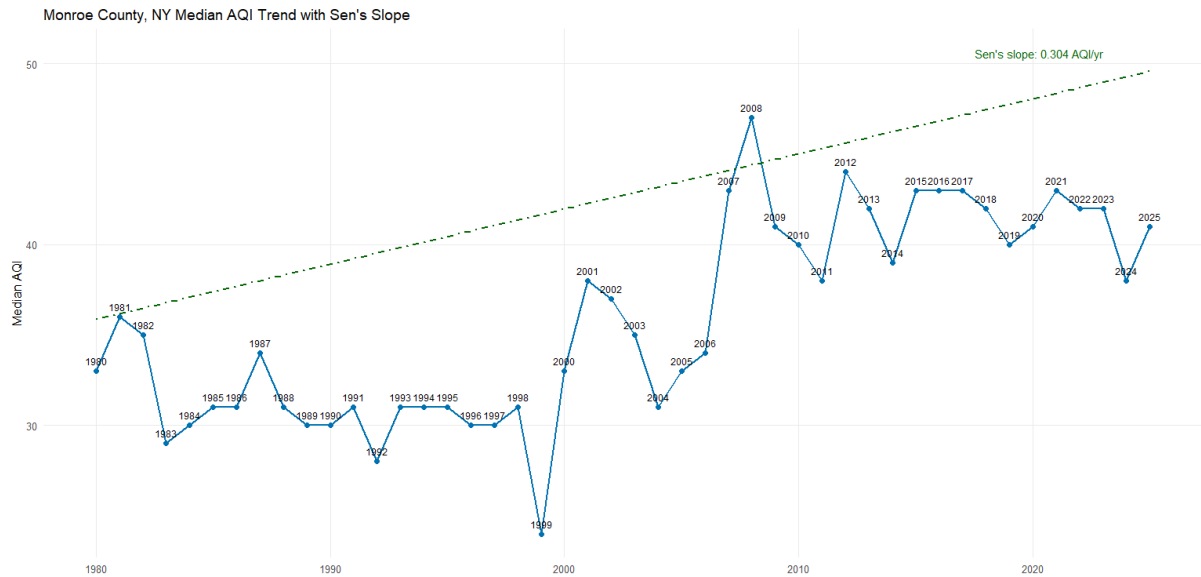


Figure 2: Annual median AQI with Sen’s slope trend estimate.

Median AQI remains relatively stable between 1980 and 1999, with values generally between 28 and 32. A structural upward shift occurs around 2000, after which median AQI fluctuates between approximately 38 and 46. Sen’s slope estimates an increase of 0.30 AQI units per year, corresponding to an approximate 13–14 unit increase over the full study period. This confirms a statistically robust long-term worsening of air quality not driven by isolated outliers.

5.3 Distributional Evidence

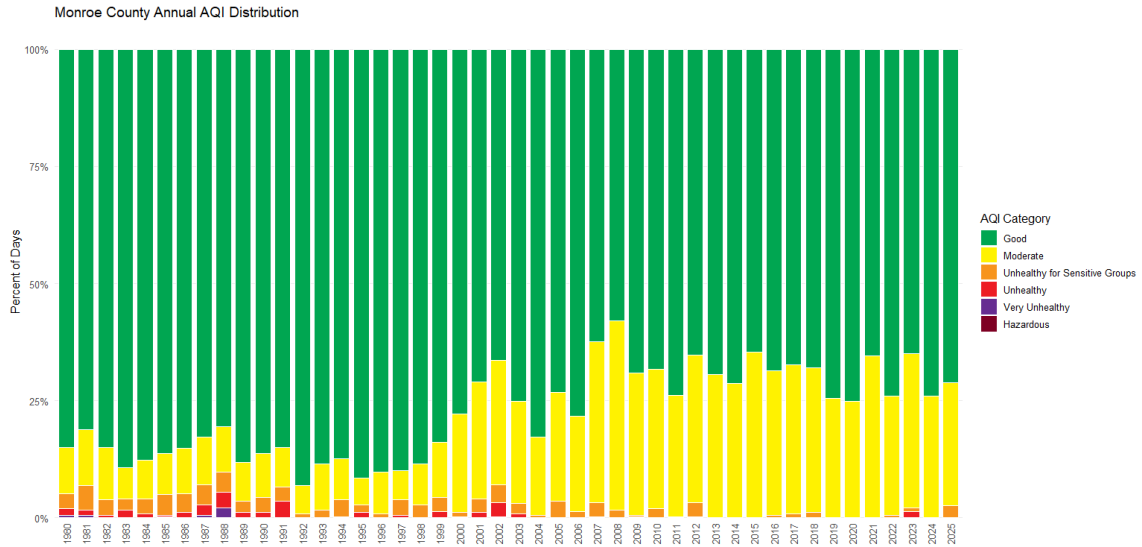


Figure 3: Stacked bar chart of annual AQI category proportions.

The stacked distribution provides clear evidence that rising median AQI is driven by frequency shifts rather than extreme events. During the 1980s and 1990s, approximately 80–90% of days were classified as “Good”. After 2000, this share declines to approximately 60–70%, while “Moderate” AQI days increase to approximately 25–35%. Unhealthy categories remain below 5% in most years.

5.4 AQI Category Distribution Shifts

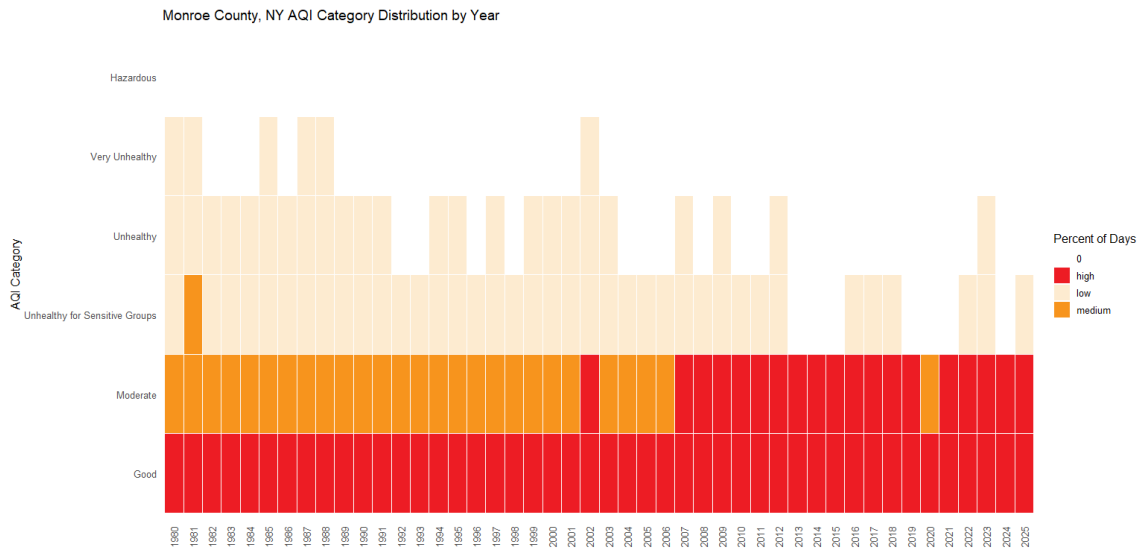


Figure 4: AQI category distribution 3-lag by Year for Monroe County (1980–2025).

6 Attributable Health Burden

The increasing prevalence of Moderate AQI exposure results in rising population attributable fractions over time. Although individual-day risk associated with Moderate AQI is lower than that of extreme pollution, the cumulative exposure affects a substantially larger portion of the population, producing a meaningful long-term health burden.

6.1 Summary Statistics

Table 1: Trend analysis results for annual median AQI.

Metric	Estimate	95% CI
Sen’s slope (Median AQI)	0.30 AQI per year	[0.21, 0.39]

6.2 Discussion

All three analytical perspectives, AQI heatmaps, median AQI trends, and stacked distributions, independently confirm a coherent pattern of chronic air quality deterioration. The findings indicate that regulatory frameworks have been effective in limiting extreme pollution events but less effective in reducing baseline pollution levels. PM_{2.5} and ozone dominate post-2000 AQI exceedances, consistent with increased urbanization, transportation emissions, and climate driven changes in atmospheric chemistry.

7 Public Health Implications

Persistent exposure to moderate air pollution levels represents a sustained population-level risk rather than an episodic hazard. In Monroe County, the data indicate that asthma morbidity is shaped by long-term exposure patterns, with disproportionate effects on children, older adults, and individuals with preexisting respiratory or cardiovascular conditions.

Even modest increases in ambient pollution are associated with measurable increases in asthma-related healthcare utilization. This reinforces the need to shift from reactive, high-AQI event response toward continuous exposure reduction strategies.

7.1 Integrated Data Interpretation and Study Context

7.1.1 Monroe County Asthma and Air Pollution Trends (2000–2023)

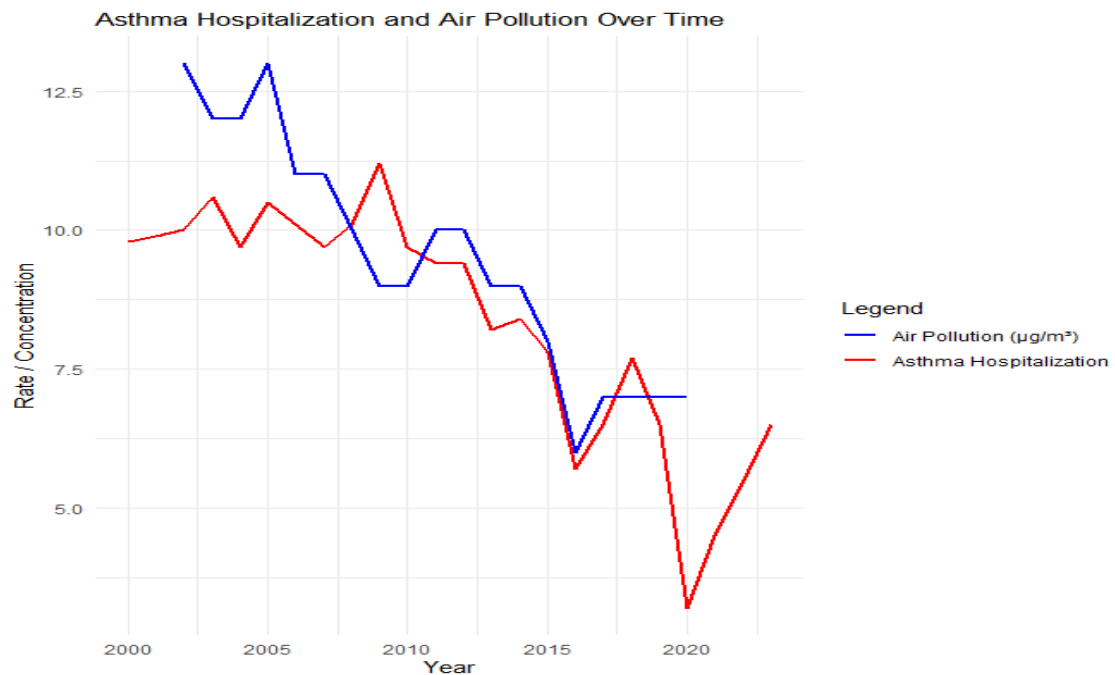
Table 2: Selected Summary of Asthma Outcomes and Air Pollution Indicators

Period	Mean Hospitalizations	Mean ED Visits	Mean Air Toxics ($\mu\text{g}/\text{m}^3$)
2000–2009	10.16	47.4	11.0
2010–2019	8.13	52.3	8.2
2020–2023	4.93	29.3	7.0

Key Observations:

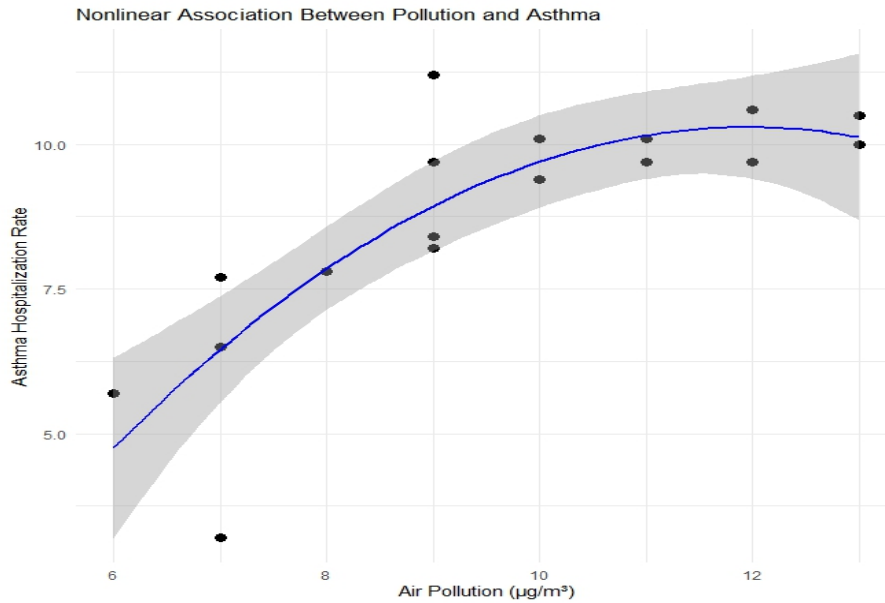
- Hospitalization rates declined over time, with a sharp drop during 2020.
- Emergency department visits remained high through 2019, indicating persistent disease burden despite declining hospitalizations.
- Air toxic concentrations decreased gradually, but not uniformly.

7.2 Visual Evidence of Nonlinear Relationships



Asthma hospitalizations and air pollution: nonlinear exposure-response relationship

7.3 Analytical Findings



Pollution effects with possible interaction

7.4 Exposure–Response Dynamics

- The relationship between air pollution and asthma hospitalizations is nonlinear.
- The steepest risk gradient occurs at low to moderate pollution levels, indicating no safe threshold.
- Model fit: Adjusted $R^2 = 0.79$, $F = 23.2$, $p < 0.001$.

7.5 Impact of System-Level Disruption (COVID-19)

- The COVID-19 period corresponds to a reduction of approximately 3.38 hospitalizations per year.
- This decline likely reflects behavioral changes, reduced exposure, and altered health-care utilization.
- Model fit improves with interaction terms: Adj $R^2 = 0.82$, $F = 41.1$, $p < 0.001$.

Asthma outcomes in Monroe County reflect the interaction of environmental exposure and systemic factors. Critically, the nonlinear exposure-response pattern indicates that most preventable harm occurs within commonly observed pollution ranges, not extreme events. 2020 suggests a short-term shock by COVID-19 produced a temporary disruption, not a structural improvement.

8 Policy-Relevant Insights

- Prioritize reduction of baseline emissions rather than focusing solely on high-AQI episodes.
- Expand real-time AQI surveillance integrated with health system alerts.
- Strengthen outpatient asthma management to reduce ED dependence.
- Incorporate nonlinear models in environmental health risk assessment.

9 Conclusion

The evidence shows that chronic exposure to moderate air pollution drives asthma morbidity more than rare extreme pollution events. While air quality improvements have occurred, they have not eliminated population risk.

You should interpret these findings as a signal to act upstream. Reducing everyday exposure levels will produce larger public health gains than episodic interventions. At the same time, maintaining healthcare system resilience during disruptions remains essential to prevent reversals in progress.

10 References

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Table 3: Integrated Asthma and Air Pollution Data, Monroe County, New York (2000–2023)

Year	Hospitalizations	ED Visits	Air Toxic Indicator	Air Toxics ($\mu g/m^3$)
2000	9.8	0.0	NA	NA
2001	9.9	0.025	NA	NA
2002	10.0	NA	13	13
2003	10.6	NA	12	12
2004	9.7	0.021	12	12
2005	10.5	55.2	0.027	13
2006	10.1	54.4	0	11
2007	9.7	52.9	0.008	11
2008	10.1	55.0	0	10
2009	11.2	59.0	0	9
2010	9.7	54.7	0.009	9
2011	9.4	55.4	0	10
2012	9.4	55.4	0	10
2013	8.2	52.7	0	9
2014	8.4	56.6	0	9
2015	7.8	52.4	0	8
2016	5.7	47.5	0	6
2017	6.5	50.5	0	7
2018	7.7	50.6	0	7
2019	6.5	47.0	0	7
2020	3.2	22.6	0	7
2021	4.5	23.8	0	NA
2022	5.5	32.6	0	NA
2023	6.5	38.3	0	NA